Prop.: V, W are finite dim. v. spaces

$T: V \to W$ $\quad A = \{\underline{v_1}, \cdots, \underline{v_n}\}$

bases: A $\quad B$ $\quad A^* = \{\phi_1, \cdots, \phi_n$

$T^*: W^* \to V^*$ $\quad B = \{\underline{w_1}, \cdots, \underline{w_m}\}$

bases: B^* $\quad A^*$ $\quad B^* = \{\psi_1, \cdots, \psi_m\}$

Then $(T_{BA})^t = T^*_{A^*B^*}$

$\downarrow$ $\quad\quad$ $\downarrow$

a $m \times n$ matrix $\quad$ a $n \times m$ matrix

Proof: Let $T_{BA} = (a_{ij})$, $T^*_{A^*B^*} = (b_{ij})$,

$T^*(\psi_j) = \sum_{r=0}^{n} b_{rj} \phi_{ij}$ $\quad$ (⇐ 证!矩阵中每一个元素都匹配到,相等?)

$\Downarrow$ (★)

$T^*(\psi_j)(\underline{v_i}) = \sum_{r=0}^{n} b_{rj} \phi_{ij}(\underline{v_i}) = b_{ij} \cdot 1$

(def of T^*) $\Longrightarrow \psi_j(T(\underline{v_i})) = b_{ij}$

$\Rightarrow \psi_j\left(\sum_{k=1}^{m} a_{ki} \underline{w_k}\right) = b_{ij}$

$\Rightarrow \sum_{k=1}^{m} a_{ki} \psi_j(\underline{w_k}) = b_{ij}$

(★) (def. of ψ_j) $\Longrightarrow$ (⇐) $a_{ji} \psi_j(\underline{w_j}) = b_{ij}$

(⇐? ⇔ ?)

$a_{ij} = b_{ij}$

Minimal Polynomials & characteristic Polynomials

Review on Polynomials

	Poly $\mathbb{F}[x]$	Integer $\mathbb{Z}$
① Basis Objects	irreducible polys., e.g. $x^2+1 \in \mathbb{R}[x]$	prime numbers 2, 3, 5, ⋯
② Unique factorization of basis objects	e.g. $x^3+1 = (x+1)(x^2-x+1)$	$20 = 2^2 \times 5$
③ Division	$\frac{p(x)}{q(x)}$ given by $p(x) = s(x)\cdot q(x) + r(x)$, remainder $\deg(r(x)) < p(x)$ (1)	

If we
make the irred. polys. $p_i(x)$
in the factorization $f(x) = a_n p_1(x)^{e_1} \cdots p_k(x)^{e_k}$
to be monic

↓

To understand $\mathbb{F}[x]$, need to find all irred. polys. in $\mathbb{F}[x]$

↓

need to find a way
to check whether a given poly. $p(x)$ has a factor $f(x)$ or not
If $p(x)$ has no factor other than itself (i.e. $p(x) = f(x)$),
then $p(x)$ is irred.

understand $\mathbb{F}(x)$ $\Rightarrow$ ~ all irred. polys. of $\mathbb{F}(x)$

$\Rightarrow$ ~ $f(x)$ of $\forall p(x) \in \mathbb{F}[x]$

$\Rightarrow$ ~ "division" of $\forall p(x) \in \mathbb{F}[x]$ by $f(x)$

$$
\begin{array}{r}
x^2 - x \\
\hline
x^2+x+1 \;\big)\; x^4 \qquad\qquad -1 \\
x^4 + x^3 + x^2 \\
\hline
-x^3 - x^2 - 1 \\
-x^3 - x^2 - x \\
\hline
x - 1
\end{array}
$$

Corollary (Root Theorem):

Let $p(x) \in \mathbb{F}[x]$,

$(x-a)$ is a factor of $p(x)$

$\Leftrightarrow$ $f(a) = 0$

Theorem (Fundamental Theorem of Algebra)

Corollary:

(1) All irred. polys in $\mathbb{C}[x]$ are of the form $\alpha x + \beta$ (i.e., of degree 1)

(2) All polys in $\mathbb{C}[x]$ are of the form $p(x) = a_n(x-\alpha_1)\cdots(x-\alpha_n)$

④ greatest common divisor

To find gcd of $(p(x), q(x))$ in $\mathbb{F}[x]$ or (a, b) in $\mathbb{Z}$ use Euclidean algorithm

gcd $(x^4-1,\ x^3-x^2)$

$x^4-1 = (x+1)(x^3-x^2) + x^2-1$

$x^3-x^2 = (x-1)(x^2-1) + (-x+1)$

$(x-1)(x+1)+0$

gcd (18, 24)

$= 6$

⑤ gcd for > 2 obj

gcd (18, 24, 36)

= gcd (gcd(18, 24), 36)

⑥ Bezout's Theorem

If gcd(x, y) = g,

then ∃ a, b, g = ax + by